

Lecture 8: Functions of Several Variables

Background & Motivation

This is where Calculus III truly begins. Until now we've worked with vectors and curves—extensions of familiar geometry. Starting today, we study functions $f(x, y)$ and $f(x, y, z)$ that take multiple inputs and return a single output. Temperature at a point in a room, pressure in the atmosphere, cost as a function of two inputs—these are all multivariate functions. We need new tools to visualize them (level curves, cross-sections) and new techniques to analyze them (partial derivatives, gradients, multiple integrals). Everything from here to the end of the course builds on this foundation.

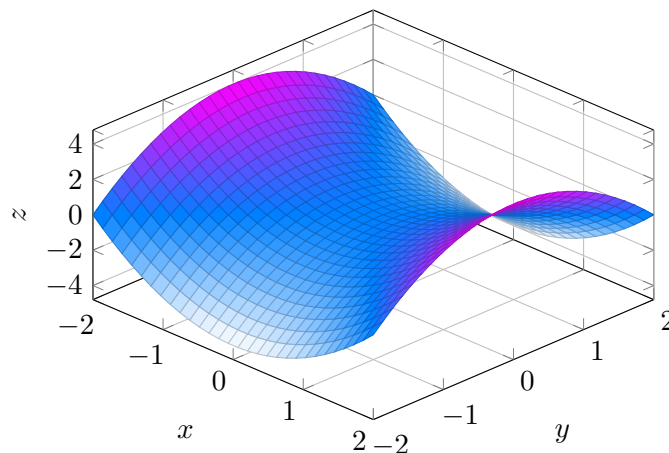
1. Functions of Two Variables

Definition 1: Function of Two Variables

A **function of two variables** $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ assigns to every point (x, y) in its domain a single real number $f(x, y)$. Whereas a single-variable function $y = f(x)$ draws a curve in the plane, a two-variable function draws a **surface** in 3D: for each input (x, y) you plot the point $(x, y, f(x, y))$, and the set of all such points is the **graph** $z = f(x, y)$. Think of the (x, y) -plane as the “floor” and the surface as a landscape whose height above that floor is determined by f .

– The graph $z = f(x, y)$ is a _____

Surface Plot of $z = x^2 - y^2$



Graph of $z = f(x, y)$ as a 3D surface. Each point (x, y) in the plane is lifted to height $f(x, y)$; here $z = x^2 - y^2$ is the classic saddle.

Example 1: Real-World Multivariable Functions

Write each as $f(\text{inputs})$:

- (a) Ohm's Law: $V = \underline{\hspace{2cm}}$ (V depends on current I and resistance R)
- (b) Kinetic energy: $KE = \underline{\hspace{2cm}}$ (depends on mass m and velocity v)
- (c) Ideal gas law: $P = \underline{\hspace{2cm}}$ (at fixed n, R : depends on T and V)

2. Domains and Ranges

Domain Restrictions to Watch For

- **Denominators** – cannot equal 0
- **Square roots** – argument ≥ 0
- **Logarithms** – argument > 0
- **Inverse trig** – $|\text{argument}| \leq 1$ (for arcsin, arccos)

Example 2: Finding Domains

Find the domain of each function:

(a) $f(x, y) = \frac{1}{x + y}$

(b) $g(x, y) = \frac{\ln(x - y)}{\sqrt{x^2 + y^2 - 4}}$

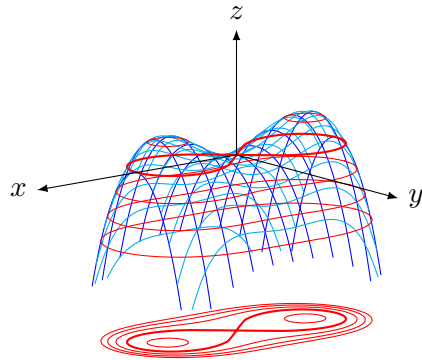
3. Level Curves

Definition 2: Level Curves

A **level curve** (or **contour**) of $f(x, y)$ at height k is the set of all points in the (x, y) -plane where f takes the value k — equivalently, the horizontal slice of the graph at height $z = k$, projected down onto the floor. Drawing many of these for different values of k produces a **topographic map** of the surface: regions where curves are packed close together correspond to *steep* parts of the surface,

while widely-spaced curves correspond to gentle slopes.

$$\{(x, y) : f(x, y) = k\}$$



A 3D surface with its level curves projected straight down onto the xy -plane. The infinity-shaped level curve is a Bernoulli lemniscate. Together, the surface and its contour projection give complementary views of the same function.

Plotting Level Curves

1. Choose several values of k in the range of f
2. For each k , set $f(x, y) = k$ and simplify
3. Identify and sketch each resulting 2D curve
4. Label each curve with its k -value

Example 3: Level Curves of a Paraboloid

Sketch level curves of $f(x, y) = x^2 + y^2$ for $k = 1, 4, 9, 16$.

Example 4: Level Curves of a Saddle

Sketch level curves of $f(x, y) = x^2 - y^2$ for $k = -4, -1, 0, 1, 4$.

4. Common 3D Equations

Surfaces You Should Recognize

Equation	Surface
$x^2 + y^2 + z^2 = r^2$	_____
$x^2 + y^2 = r^2$	_____
$z = x^2 + y^2$	_____
$x^2 + y^2 = z^2$	_____
$z = x^2 - y^2$	_____
$Ax + By + Cz = D$	_____

5. Functions of Three or More Variables

Definition 3: Level Surfaces

For $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, we cannot graph $w = f(x, y, z)$ (requires 4D). Instead we use **level surfaces**:

$$\{(x, y, z) : f(x, y, z) = k\}$$

– The 3D analog of level curves

Example 5: Level Surfaces of $x^2 + y^2 + z^2$

Describe the level surfaces of $f(x, y, z) = x^2 + y^2 + z^2$.

Practice Problems

Problem 1. Find the domain of $f(x, y) = \sqrt{9 - x^2 - y^2}$.

Problem 2. Find the domain and range of $f(x, y) = e^{-(x^2 + y^2)}$.

Problem 3. Sketch level curves of $f(x, y) = xy$ for $k = -2, -1, 0, 1, 2$.

Problem 4. Describe the level surfaces of $T(x, y, z) = 100 - x^2 - y^2 - z^2$.

Problem 5. The wind chill index is $W(v, T) = 35.74 + 0.6215T - 35.75v^{0.16} + 0.4275Tv^{0.16}$. Find W at $T = 0^\circ\text{F}$ and $v = 25$ mph. What do the level curves represent physically?